DBMS FOR SOCIAL MEDIA INTERACTIONS

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INTRODUCTION TO CHOSEN PROBLEM

Managing Large-Scale Social Media Data

- Social media platforms generate massive amounts of interaction data.
- Users continuously create posts, comments, likes, and other activities.
- These interactions produce large relational datasets.
- Efficient storage and retrieval of this data requires a Database Management System (DBMS).
- As user activity increases, query processing on this data becomes computationally expensive.

INTRODUCTION TO CHOSEN PROBLEM

Challenges in Processing Social Media Data

- Social media databases contain millions of records.
- Operations such as filtering, aggregation, sorting, and joins become slow on large datasets.
- Traditional databases rely mainly on CPU-based query processing.
- CPUs have limited parallel processing capability.
- This leads to slower query execution for large-scale data analysis.

BACKGROUND AND MOTIVATION

Database Systems for Social Media Platforms

- Social media systems rely on relational databases to store structured interaction data.
- Core entities include Users, Posts, Likes, Comments, and UserActivity.
- Relationships between these entities represent user interactions on the platform.
- Efficient database design allows structured storage and consistent data management.

BACKGROUND AND MOTIVATION

Social media platforms store huge amounts of data including posts, likes, comments, and messages.

- Over time, these platforms accumulate millions or even billions of records.
- When users try to search for an old message, comment, or post, the system must scan large datasets.
- As the data grows, traditional CPU-based query processing can become slow.
- Faster database processing techniques are required to maintain good user experience.
- GPU-based parallel processing can help accelerate query execution on large datasets.

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Traditional DBMS struggle with slow query execution when processing large social media interaction datasets.

Objectives

- Design a relational database for social media interactions
- Model entities such as Users, Posts, Likes, Comments, and UserActivity
- Explore GPU acceleration for database query processing
- Compare performance between CPU and GPU execution

GENERAL SOLUTION

Database Design for Social Media Interactions

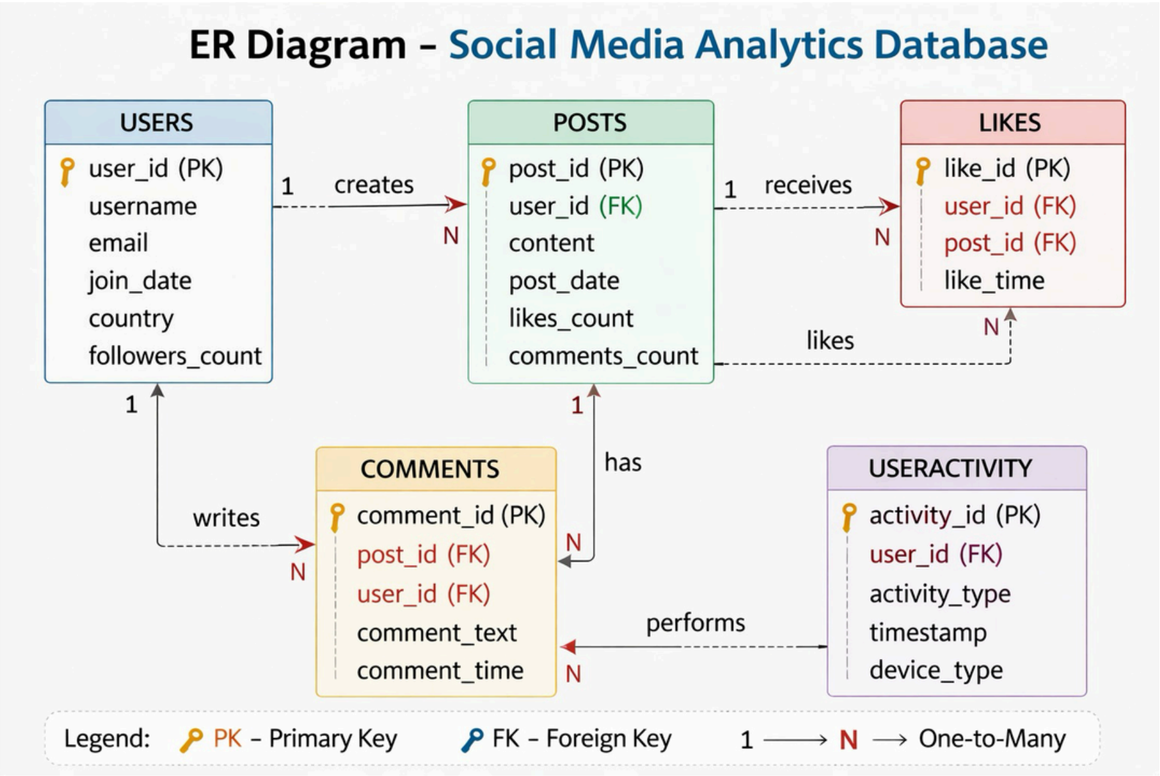
The proposed system models social media interactions using a relational database.

Key entities include:

- Users – stores information about registered users
- Posts – stores content created by users
- Likes – records when users like posts
- Comments – stores comments made on posts
- UserActivity – logs user actions on the platform

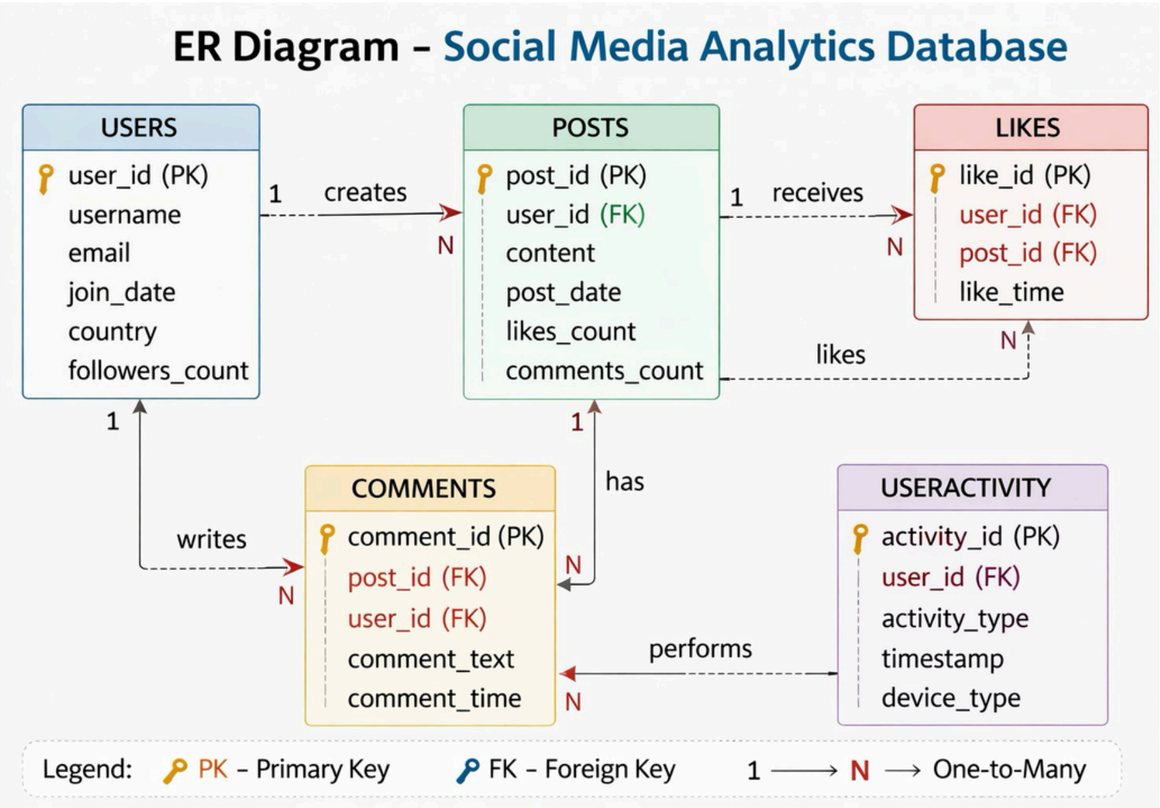
These entities are connected through primary key and foreign key relationships.

GENERAL SOLUTION



- The ER diagram represents the structure of the social media database.
- It shows relationships between Users, Posts, Likes, Comments, and UserActivity.
- One user can create multiple posts.
- A post can receive likes and comments from many users.
- The database structure captures user interactions efficiently

DATABASE STRUCTURE



Database Entities (Tables)

- The database is organized into relational tables connected using primary and foreign keys.
- Main entities: Users, Posts, Likes, Comments, UserActivity.

Table Overview

- Users – Stores user details (user ID, username, email, join date).
- Posts – Stores posts created by users (content, time, engagement).
- Likes – Records when a user likes a post (many-to-many relation).
- Comments – Stores comments made by users on posts.
- UserActivity – Logs user actions like login, posting, liking, and commenting.

PLAN OF IMPLEMENTATION

Implementation Steps

- Design relational database schema for social media interactions
- Create tables for Users, Posts, Likes, Comments, and UserActivity
- Load datasets representing social media interactions
- Implement query operations such as filtering, aggregation, sorting, and joins
- Use GPU-based processing to accelerate these operations
- Compare CPU and GPU query execution performance

DBMS RELATED SOLUTION: WHY?

Why Use a Database Management System?

- Social media platforms generate large volumes of structured interaction data.
- Data such as users, posts, likes, and comments must be stored in an organized manner.
- A DBMS allows efficient storage, retrieval, and management of this data.
- Relational databases help maintain relationships between entities using primary and foreign keys.
- DBMS enables efficient query processing for analyzing social media interactions.

DBMS RELATED SOLUTION: WHY?

Why GPU for DBMS?

- Database operations are highly parallelizable
  - GPUs handle large-scale computations efficiently
  - Faster processing of analytical queries
  - Improved performance for big datasets
  - Useful for real-time analytics systems

EXPECTED OUTPUT

- A working relational database model for social media interactions
- Efficient representation of user activities and relationships
- Execution of analytical queries on large datasets
- Demonstration of GPU acceleration for database operations
- Performance comparison showing faster query execution using GPUs

THANK YOU